

SAMAYUKTAM

Sorting Benchmark Report

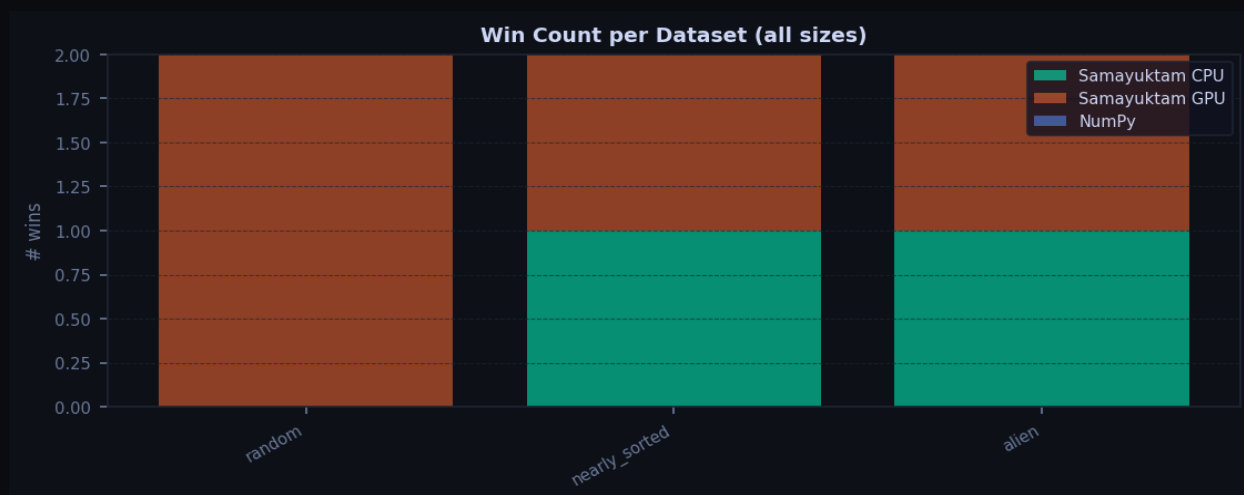
Generated: 2026-05-31 10:36

Float32 · NaN excluded · -0.0 excluded · Best-of-N runs

- CPU (/mnt/c/Users/Swapnopam/Desktop/Prog/CAGM Improved/huggingfacepage/shit/test/samayuktam_cpu)
- GPU (/mnt/c/Users/Swapnopam/Desktop/Prog/CAGM Improved/huggingfacepage/shit/test/samayuktam_gpu)

Metric	Value
CPU Best Time (random, largest)	179.86 ms
CPU Throughput	2121 MB/s
CPU Speedup vs NumPy Quick	4.47×
GPU Best Time (random, largest)	167.91 ms
GPU Throughput	2272 MB/s
GPU Speedup vs NumPy Quick	4.54×
GPU / CPU Speedup	1.07×
Array sizes tested	10M, 100M
Dataset types	3

Win Count by Dataset — which algorithm sorted fastest most often



Dataset: Random

Uniformly random float32 values in $[-1e9, 1e9]$.



Raw Numbers

Size	Sam CPU (ms)	CPU MB/\$	Sam GPU (ms)	GPU MB/\$	NP Quick (ms)	NP Merge (ms)	NP Stable (ms)	Spdup CPU/Q	Spdup GPU/Q
10M	16.65	2292	16.61	2297	83.84	937.75	923.60	5.04×	5.05×
100M	179.86	2121	167.91	2272	804.35	11024.47	10818.74	4.47×	4.79×

Dataset: Nearly Sorted

Sorted with 1% of elements randomly swapped.



Raw Numbers

Size	Sam CPU (ms)	CPU MB/s	Sam GPU (ms)	GPU MB/s	NP Quick (ms)	NP Merge (ms)	NP Stable (ms)	Spdup CPU/Q	Spdup GPU/Q
10M	48.09	793	45.31	842	69.07	194.14	181.89	1.44×	1.52×
100M	132.54	2878	139.13	2742	809.03	2328.35	2336.66	6.10×	5.81×

Dataset: Alien

Mixed extremes: subnormals, huge-magnitude floats, tight clusters.



Raw Numbers

Size	Sam CPU (ms)	CPU MB/\$	Sam GPU (ms)	GPU MB/\$	NP Quick (ms)	NP Merge (ms)	NP Stable (ms)	Spdup CPU/Q	Spdup GPU/Q
10M	52.61	725	43.23	883	62.30	918.49	913.00	1.18×	1.44×
100M	199.32	1914	206.20	1850	706.67	10421.92	10300.25	3.55×	3.43×

Methodology

Parameter	Detail
Data type	float32 (4 bytes/element), little-endian binary files
Exclusions	NaN values removed; -0.0 flipped to +0.0 before writing
Timing	Best-of-N wall-clock time via <code>time.perf_counter()</code> (numpy) / binary-internal timer
Throughput	MB/s = $(n \times 4) / (1024^2 \times \text{seconds})$
NumPy Quick	<code>np.sort(kind='quicksort')</code> — introsort hybrid
NumPy Merge	<code>np.sort(kind='mergesort')</code> — stable merge sort
NumPy Stable	<code>np.sort(kind='stable')</code> — Timsort
Samayuktam	Custom radix/parallel sort; binary fed via stdin with path

Dataset Descriptions

random — Uniformly random float32 values in $[-1e9, 1e9]$.

nearly_sorted — Sorted with 1% of elements randomly swapped.

alien — Mixed extremes: subnormals, huge-magnitude floats, tight clusters.